



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 10 • STANDARD 6.GR.4

Surface Area of Pyramids

Lesson 10-5 • Teacher Copy

Name: _____ **Date:** _____ **Class:** _____



TODAY'S OBJECTIVES

Content Objective: I can find the surface area of a pyramid by adding the base area and the lateral faces.

Language Objective: I can explain my work using the words pyramid, slant height, lateral face, and base.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Pyramid

Slant height

Lateral face

Base

Apex

Lateral area



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

A solid with a polygon base and triangular faces meeting at one point is a

2

The height of a triangular face of a pyramid, measured along the slanted side, is the

3

A triangular side face of a pyramid is a

4

The polygon on the bottom of a pyramid is the

5

The single top point where the triangular faces of a pyramid meet is the

6

The combined area of all the side faces, not counting the base, is the

Reflect — Exit Ticket

A square pyramid has a base edge of 6 in and a slant height of 5 in. What is the total surface area?

A. 96 in^2

B. 60 in^2

C. 96 in^3

D. 36 in^2

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Pyramid
2. **Notes 2:** Slant height
3. **Notes 3:** Lateral face
4. **Notes 4:** Base
5. **Notes 5:** Apex
6. **Notes 6:** Lateral area
7. **Watch for this mistake:** *A common mistake in Surface Area of Pyramids is skipping the key idea: "Surface area of a pyramid = base area + the area of all the triangular lateral faces." — always check your work against this rule before you submit.*
8. **Try It 1:** A. 20 in^2 — *Area of a triangle = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 8 \times 5 = 20 \text{ in}^2$. You use the slant height (5) as the triangle's height, not the pyramid's vertical height.*
9. **Try It 2:** A. 81 in^2 — *The base is a square, so base area = side $\times$ side = $9 \times 9 = 81 \text{ in}^2$. You would still add the four triangular faces to get the full surface area.*
10. **Exit Ticket:** A. 96 in^2 — *Base: $6 \times 6 = 36 \text{ in}^2$. Each lateral face: $\frac{1}{2} \times 6 \times 5 = 15 \text{ in}^2$. Four faces: $4 \times 15 = 60 \text{ in}^2$. $SA = 36 + 60 = 96 \text{ in}^2$. Surface area uses square units (in^2).*